

[VPP-6] Data Validation Rules: VEE Validation Checks Business Requirement

Data Validation Business Requirements Document (BRD)

Document Status	NEW
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Document Name	Data Validation Business Requirements based on SQMD guidelines
References	• Direct Access Standards for Metering and Meter Data (DASMMD) • https://growingenergylabs.atlassian.net/wiki/x/PABylgI Request access
Applicable Systems	• Solar inverter telemetry platforms • Battery sub-meter telemetry platforms • All OEM pipeline data
Expansion of this BRD	• Historical Data Quality Examples outlining how these data validation issues could be preventable/detectable by following procedures below: Historical Data Quality Issues & Validation Control Mapping • An expansion on data estimation rules + procedures: Data Estimation & Editing Procedure S

Background & Purpose

Accurate, settlement-quality telemetry is foundational to Qcells' participation in grid services programs, incentive programs, and performance-based reporting.

To ensure consistent data integrity, auditability, and reuse across programs, Qcells applies standardized data validation rules aligned with CPUC Settlement Quality Meter Data (SQMD) principles, which represent the prevailing metering and data quality standard for all of the US.

This BRD defines the business requirements for data validation only. Separate standards govern estimation and editing.

2. Scope

In Scope for Data Validation:

- Interval-based solar production telemetry
 - Interval-based battery operational telemetry
 - Data used for:
 - Grid services programs
 - Settlement and incentive calculations
 - Performance analytics and forecasting
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3. Guiding Standards

The data validation rules defined in this BRD are based on:

- CPUC Settlement Quality Metering Data (SQMD) principles
- DASMM validation practices
- IOU-accepted data quality standards

(As listed above in References)

4. Success Criteria:

Data Validation Goals	Related Data Validation Check
All data is attributable to a known asset and site	Channel mapping, Meter Identity
All intervals are temporally accurate and complete	Interval Completeness, Time Integrity
All values are physically and statistically plausible	Spike Detection, High/Low Usage

All validation outcomes are traceable and auditable	Sum Check
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5. Data Inputs

Data Type	Description
Solar Telemetry	Inverter-level production data
Battery Telemetry	Sub-metered charge/discharge data/SoC
Grid Import/Export Data	Received from OEM APIs
Consumption Data	Calculated field from OEM/Qcells data
Enrollment Metadata	Asset ID, site ID, capacity, etc
Time Reference	Timezone-aware system time

6. Required Validation Checks (Summary)

Validation Check	Technical Summary	Minimally Acceptable Method	Estimated Difficulty to Implement
Interval Completeness	Verify presence of all expected 15-minute intervals (96 per day, DST-adjusted)	Perform a quick check to check that are 96 intervals per day per site ID. Flag if not equal to 96.	Easy
Sum Check	Verify internal consistency across channels	Compare interval-summed energy with	Easy

	and scaling factors	expected total energy.	
High / Low Usage	Compare interval values against historical distributions	Excel check: compare against system nameplate or historical max/min using	TBD
Spike Detection	Identify single-interval deviations exceeding statistically reasonable bounds	If higher output than nameplate capacity, flag as suspect	Easy
Channel Mapping	Ensures telemetry channels are currently mapped to physical power flows (charge vs. charge, import vs. export)	Cross-check telemetry channels	TBD
Time Integrity	Validate interval timestamps align correctly within event windows + temporal reporting boundaries	Sort intervals by timestamps and confirm 15-minute increments using =A2-A1 check. Should equal 00:15. Flag if intervals deviate from 15.	Medium

Meter Identity	Confirm meter/service account mapping has not changed	Compare telemetry meter ID/device ID and site ID against enrollment list. Use vlookup/xlookup.	Easy
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7. Validation Rules (Detailed Requirements)

7.1 Interval Completeness Validation

Rule Objective	Rule Definition
Ensure all expected settlement or reporting intervals are present and accounted for.	For each asset and calendar day, the system shall verify the presence of all expected intervals based on: • Configured interval length • Daylight Saving Time adjustments Expected interval counts must be dynamically calculated and not assumed to be constant.

Validation Criteria	Failure Conditions
All intervals are present for the applicable day type	Missing intervals not attributable to DST
Intervals aligned to the configured time grid	Duplicate intervals without unique temporal identifiers
Duplicate or missing intervals are explicitly identified	Misaligned interval boundaries

Failure Scenario Examples:

- A daily telemetry set contains fewer than the expected number of 15-minute intervals after accounting for Daylight Saving Time adjustments.
- One missing interval is silently ignored, causing daily energy totals to understate actual production or dispatch performance.

7.2 Sum and Internal Consistency Validation

Rule Objective	Rule Definition
Ensure mathematical consistency across reported values, channels, and units.	Reported power, energy, and net values must reconcile according to physical and mathematical relationships.

Validation Criteria	Failure Conditions
Energy equals power multiplied by interval duration	kWh values inconsistent with kW and interval length
Net values equal the sum of component channels	Net totals not reconcilable with components
Scaling factors and unit conversions are consistent	Scaling or unit mismatches

Failure Scenario Examples:

- Reported interval energy does not equal reported power multiplied by the interval duration.
- Scaling errors cause aggregated kWh values to diverge from summed kW intervals, leading to incorrect reporting and settlement disputes.

7.3 High / Low Usage Validation

Rule Objective	Rule Definition
Detect sustained values that are statistically or physically implausible	Intervals shall be evaluated against: • Asset nameplate and operational constraints

	• Historical distributions for the same asset and time-of-day This rule targets prolonged anomalies rather than isolated spikes.
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Validation Criteria	Failure Conditions
Output remains within physical limits	Prolonged output exceeding capacity constraints
Sustained behavior aligns with historical patterns	Extended flatlining or constant values indicative of sensor failure
	Sustained deviations outside historical bounds

Failure Scenario Examples:

- Sustained interval values remain within absolute capacity limits but fall outside historical distributions for the same asset and time-of-day.
- A sensor becomes stuck at a constant output level, masking true operational behavior and overstating performance over time.

7.4 Spike Detection Validation

Rule Objective	Rule Definition
Identify isolated interval values that are not representative of real system behavior	Single-interval deviations shall be evaluated relative to: • Adjacent interval values • Statistically derived bounds based on historical behavior • Asset-specific ramp-rate and capacity constraints Spikes are defined as abrupt, non-sustained deviations that

	immediately return to baseline behavior.
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Validation Criteria	Failure Conditions
Interval-to-interval changes fall within plausible ramp limits	Single-interval excursions significantly exceeding expected variability
Values remain within statistically reasonable deviation thresholds	Output changes are inconsistent with known asset response characteristics

Failure Scenario Examples:

- A single 15-minute interval reports power output that exceeds the asset’s historical statistical bounds while adjacent intervals remain within normal operating ranges.
- A transient telemetry glitch produces one extreme interval value that inflates aggregated production totals if it is not flagged and excluded.

7.5 Channel Mapping Validation

Rule Objective	Rule Definition
Ensure that all reported power and energy values are associated with the correct physical flow and channel definition.	Each telemetry channel must map unambiguously to a defined physical meaning: • Solar: production/export channels represent the generated power • Battery: charge and discharge channels represent distinct operational states Net power calculations must use consistent sign conventions and channel-aggregation logic.

Validation Criteria	Failure Conditions
Channel definitions match enrollment and configuration records	Inverted sign conventions
Charge, discharge, import, and export are not inverted or co-mingled	Charge data reported as discharge (or vice versa)
Net values reconcile correctly to component channels	Export/import channels swapped or netted incorrectly

Failure Scenario Examples:

- A battery inverter reports discharge power on a charge-designated telemetry channel due to incorrect channel configuration, resulting in an inverted net power sign.
- Battery discharge is recorded as charging because charge and discharge channels are swapped, causing downstream systems to treat real output as non-performance.

7.6 Time Integrity Validation

Rule Objective	Rule Definition
Ensure interval timestamps accurately reflect real elapsed time and remain aligned with system time references.	Telemetry timestamps shall be validated against a trusted time reference to detect excessive clock drift.

Validation Criteria	Failure Conditions
Timestamps are timezone-aware and include UTC offset	Drift exceeding the accepted tolerance thresholds (to be defined)
Clock drift remains within the to be defined tolerance	Intervals shifted outside expected temporal windows
Temporal ordering of intervals is consistent	Inconsistent or unstable clock behavior

Failure Scenario Examples:

- Interval timestamps drift from system time by more than the accepted tolerance, resulting in misalignment with defined temporal boundaries.
 - Valid discharge occurs during an event window but is timestamped earlier due to device clock drift, leading to disallowed or misattributed performance.
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7.7 Meter Identity Validation

Rule Objective	Rule Definition
Ensure telemetry is attributed to the correct physical asset and service location	Each interval must be associated with: • A valid site or service account (definition may be specified by Ops/Dev team)

Validation Criteria	Failure Conditions
Device identifiers match enrollment records	Unknown or mismatched device identifiers
Telemetry timestamps fall within authorized operational windows	Conflicting device-to-site mappings
No unauthorized device substitutions occur	

Failure Scenario Examples:

- Telemetry is received from a device identifier that does not match the enrolled asset mapped to the service account.
 - A device replacement occurs without updating enrollment records, causing data to be attributed to the wrong asset.
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8. Validation Outcomes Requirements:

Each interval should be assigned a validation status indicating:

- Valid (eligible for downstream use)
- Suspect (requires estimation or review)
- Invalid (excluded from use)

Notes on Auditability:

The validation framework must support:

- Interval-level traceability
- Reproducibility of validation outcomes
- Historical retention of validation flags and reasons
- Sampling-based audits by internal or external parties

This is the standard required for nearly all VPP programs.

9. Timeline & Milestones

Milestone	Date	
Data Validation methods are actively in place	May 1, 2025	

10. Stakeholders

Team	Responsibility	Contact
G&ES Dev	• Implementing these validation techniques	@Naveen Siddalingaswamy @Sannjeev Lakkaraju
G&ES Program	• BRD • Process Examples & Clarifications	@Alexandra Clark @Dongmin Son
G&ES Product	• Product Delivery	@Cecilia Zhou

G&ES Ops	• Added improvements/notes on what will work best for Dev team	@Shuai Zhang @Kaifeng Xu
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